Classic Techniques

Arman Ferdous

Prefix Sums

- Discussed in the last class
- Main idea:

$$a[L] + \ldots + a[R] = (a[1] + \ldots + a[R]) - (a[1] + \ldots + a[L - 1])$$
$$= pre[R] - pre[L - 1]$$

  With O(n) preprocess to build pre[], we can query in O(1)
- What other type of operations let us use similar trick?
  Multiplication, min/max, gcd, lcm, and, or, xor…

Difference Array

- For an array a[1…n], we call its difference array d[2…n] as

$$d[i] = a[i] - a[i - 1]$$

  d[i] is the amount of change from the previous index a[i] = a[i - 1] + d[i]
- Question: If we know d[2…n], can we retrieve a[1…n] ?
- Question: How do we retrieve a[1…n]?

Relationship between prefix sums and difference array

- If we know pre[1…n], can we figure out a[1…n]?
- If we think of taking prefix sum of an array as an operator, and taking difference array of an array as another operator, then they act inversely from each other.

Application of difference array

- Suppose you have an array of n elements, all 0 initially.
- There will be q updates to this array. The ith update will add 1 to cells L_i to R_i.
- You have to print the final state of the array in the end.

- Notice, we can’t do update and query simultaneously. The query must be done at the end for this trick to work.

https://codeforces.com/contest/816/problem/B

Karen, a coffee aficionado, wants to know the optimal temperature for brewing the perfect cup of coffee. Indeed, she has spent some time reading several recipe books, including the universally acclaimed "The Art of the Covfefe".

She knows n coffee recipes. The i-th recipe suggests that coffee should be brewed between l_i and r_i degrees, inclusive, to achieve the optimal taste.

Karen thinks that a temperature is *admissible* if at least k recipes recommend it.

Karen has a rather fickle mind, and so she asks q questions. In each question, given that she only wants to prepare coffee with a temperature between a and b, inclusive, can you tell her how many admissible integer temperatures fall within the range?

Input

The first line of input contains three integers, n, k ($1 \le k \le n \le 200000$), and q ($1 \le q \le 200000$), the number of recipes, the minimum number of recipes a certain temperature must be recommended by to be admissible, and the number of questions Karen has, respectively.

The next n lines describe the recipes. Specifically, the i-th line among these contains two integers l_i and r_i ($1 \le l_i \le r_i \le 200000$), describing that the i-th recipe suggests that the coffee be brewed between l_i and r_i degrees, inclusive.

The next q lines describe the questions. Each of these lines contains a and b, ($1 \le a \le b \le 200000$), describing that she wants to know the number of admissible integer temperatures between a and b degrees, inclusive.

Output

For each question, output a single integer on a line by itself, the number of admissible integer temperatures between a and b degrees, inclusive.

Previous Smaller Element

- Given an array of integers a[1…n]. For each index i, we wish to find the largest j < i such that a[j] < a[i].
- Naive solution?

- We can use the same technique to solve (Previous/Next) (Smaller/Greater) element problem.

Alive & Dead elements

- When an element never gets to be a pse again?
- How the alive elements look.
- Formulate the solution.
- Complexity analysis.

Pseudocode

```
let A = array with indices 1 to n
let F = array with indices 1 to n, for keeping answers
let S = empty stack

for i from 1 to n:
        while not S.empty() and A[S.top()] >= A[i]:
                S.pop()

        if S.empty(): F[i] = 0
        else: F[i] = S.top()

        S.push(i)
```

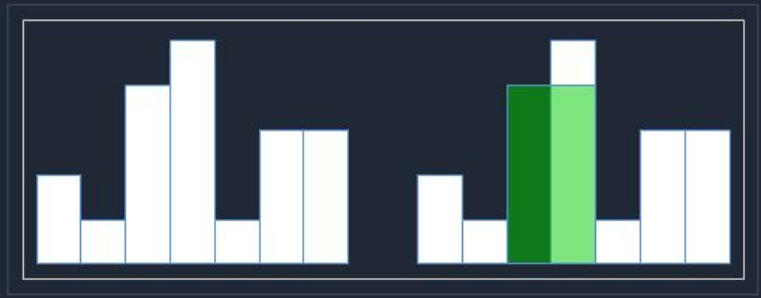
Nearest Smaller element

- For each index i of an array, find j such that

$$a[j] < a[i] \quad \text{and} \quad |i - j| \text{ is smallest}$$

https://lightoj.com/problem/histogram

A histogram is a polygon composed of a sequence of rectangles aligned at a common base line. The rectangles have equal widths but may have different heights. For example, the figure shows the histogram that consists of rectangles with the heights `{2, 1, 4, 5, 1, 3, 3}` measured in units where the width of the rectangles is 1.



Usually, histograms are used to represent discrete distributions, e.g., the frequencies of characters in texts. Note that the order of the rectangles, i.e., their heights, is important. Calculate the area of the largest rectangle in a histogram that is aligned at the common base line, too. The figure on the right shows the largest aligned rectangle for the depicted histogram.

Input

Input starts with an integer **T (≤ 20)**, denoting the number of test cases.

Each case contains a line with an integer **N (1 ≤ N ≤ 30000)** denoting the number of rectangles. The next line contains **N** space separated positive integers **(≤ 30000)** denoting the heights.

http://poj.org/problem?id=3494

Given a m-by-n (0,1)-matrix, of all its submatrices of all 1's which is the largest? By *largest* we mean that the submatrix has the most elements.

Input

The input contains multiple test cases. Each test case begins with m and n ($1 \le m, n \le 2000$) on line. Then come the elements of a (0,1)-matrix in row-major order on m lines each with n numbers. The input ends once EOF is met.

Output

For each test case, output one line containing the number of elements of the largest submatrix of all 1's. If the given matrix is of all 0's, output 0.

Sample Input

```
2 2
0 0
0 0
4 4
0 0 0 0
0 1 1 0
0 1 1 0
0 0 0 0
```

Sample Output

```
0
4
```

All subarray sum

- Given an array of n integers, let f(L, R) be a[L] + … + a[R].
- Compute the following sum

$$\sum_{1 \le L \le R \le n} f(L, R)$$

Sum of all subarray minimums

- Given an array of n integers, let f(L, R) be min(a[L], …, a[R])
- Compute the following sum

$$\sum_{1 \le L \le R \le n} f(L, R)$$